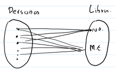
Est SEP 28

lunes, septiembre 28, 2026 2:18 PM

Relaciones y Funciones

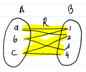
Ej:



Df: Una Relación R es un subjunto de $A \times B$. Entre elementos de A y un conjunto B.

Ej: R = "leyó el libro" = $\{(p, \ell) : p$ persona, ℓ libro, p leyó a $\ell\}$.

Ej:

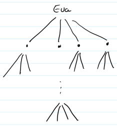


$R = \{(a,1), (a,2), (b,1), (b,3), (c,2), (c,4)\}$.

La relación puede ocurrir dentro del mismo conjunto A.

$R \subseteq A \times A$.

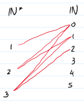
Ej: Ser madre:



Ej: $R = \bigcup_{n \in \mathbb{N}^*} = \{(\{n\} \times \{0,1,2,\dots,n-1\})\}$

$= \{\underbrace{\{1\} \times \{0\}}_{n=1}\} \cup \{\underbrace{\{2\} \times \{0,1\}}_{n=2}\} \cup \{\underbrace{\{3\} \times \{0,1,2\}}_{n=3}\} \cup \dots$

$= \{\underbrace{(1,0)}_{n=1}, \underbrace{(2,0), (2,1)}_{n=2}, \underbrace{(3,0), (3,1), (3,2)}_{3=n}, \dots\}$



R = Ser mayor que.

R = los residuos = %

Df: Dada R una relación:

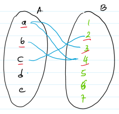
$\mathrm{Dom}(R) = \{x : x \in A \wedge \exists y \text{ tq } (x,y) \in R\}$

Ej: $= \{a, b, c\}$

$\mathrm{Im}(R) = \{y \in B : \exists x \text{ tq } (x,y) \in R\}$

$= \{2, 3, 4\}$

$\mathrm{Cmp}(R) = \{a, b, c, 2, 3, 4\}$



(π^2, π)

Ej: $R = \{(x,y) \in \mathbb{R}^2 : y^2 = x\}$

$y^2 = x$
$16 = 2$

$y^2 = \frac{1}{16} \not< \frac{1}{2} = x$

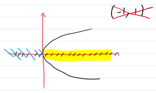
Elementos en R: $(0,0)$, $(1,1)$, $(1,-1)$, $\cancel{(2,4)}$, $\cancel{(\frac{1}{2}, \frac{1}{4})}$, $(\frac{1}{4}, \frac{1}{2})$. $(2, \sqrt{2})$

$(4,2)$.

$\mathrm{Dom}[R] = \mathbb{R}^{\geq 0}$

$\cancel{(-1,1)}$

$(\sqrt{2}, \sqrt{\sqrt{2}}) = (2^{1/2}, 2^{1/4})$



$\cancel{(-1,i)}$

$\mathrm{Ran}[R] = \mathbb{R}$

Ej: $R = \{(x,y) \in \mathbb{R}^2 : xy = 1\}$

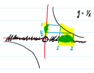
$(\frac{1}{2}, 2)$

$(\pi, \frac{1}{\pi})$

$\mathrm{Dom}[R] = \mathbb{R}^*$

$\downarrow$

$y = 1/x$



A = intervalo = $[1,2]$

$R[A] = [\frac{1}{2}, 1]$.

Teorema: R sea una relación, $A \subseteq \mathrm{Dom}(R)$

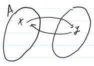
$A \subseteq R^{-1}[R[A]]$

Prueba: Sea $x \in A$, $\left.\begin{matrix} x \sim y \checkmark \\ (x,y) \in R \checkmark \\ \boxed{R(x) = y} \end{matrix}\right\}$

$\Rightarrow x \in \mathrm{Dom}(R) \Rightarrow (x,y) \in R$ para algún y.

$y \in R^{-1}[A]$

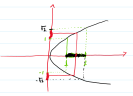
$x \underset{R}{\sim} y$
$x \sim y$



$\Rightarrow x \sim y \quad x \in R[R^{-1}[A]]$.

Ej: $R = \{(x,y) \in \mathbb{R}^2 : y^2 = x\}$.

$A = [1,2] \qquad R[A] = [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$.

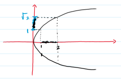


$2 = (\sqrt{2})^2$
$x = y^2$

$(1,1)$, $(1,-1)$
$(2,\sqrt{2})$ $(2,-\sqrt{2})$

$B = [1, \sqrt{2}]$

$R^{-1}[B] = [1,2]$



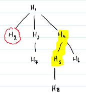
$A = [1,2]$
$R[A] = [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$
$R^{-1}[R[A]] = [1,2]$.

$A = R^{-1}[R[A]]$

¿Siempre son iguales?

$\downarrow$

Diga una R, A tal que: $A \subsetneq R^{-1}[R[A]]$



R = "Es padre de"

$A = \{\cancel{H_2}, H_3, H_4\}$

$R[A] = \{H_5, H_6, H_7\}$

R^{-1} = "Es hijo de".

$A \subseteq R^{-1}[R[A]] = \{H_4, H_3\}$ ✓

$H_5 \underset{R^{-1}}{\sim} H_4$

Df: Sea $R \subseteq A \times B$ una relación:

definimos la relación inversa $R^{-1} = \{(y,x) : (x,y) \in R\}$

Ej: R = "Ser padre de"

R^{-1} = "Ser hijo de".

R = "Ser hermano de"

$R^{-1} = R$ Se dice que R simétrica.

Df: R es una relación simétrica si $(x,y) \in R \Rightarrow (y,x) \in R$.

Ej: R_1 = "Ser madre de" No es simétrica.

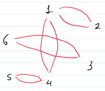
R_2 = "Amar a" No

R_3 = "Ser amigo de" ← Depende de "amigo".

R_4 = "Ser compañero de clase de Combariza".

R_5 = "Ser igual a".

R_6 =



$= \{(1,2), (2,1), (1,4), (4,1)$
$(3,6)\ (6,3), (4,5)\ (5,4)\}$

$R_7 = \{(z, -z) : z \in \mathbb{Z}\}$.

$R_8 = \{(x,y) \in \mathbb{R}^2 : |x| = |y|\}$.